FINANCE, GROWTH AND DECAY TEST

Instructions

- Answer all questions.
- Show all necessary calculations and formula substitutions.
- Give interest rates correct to two decimal places where necessary.
- This is a closed-book test. Using your notes is not allowed.

QUESTION 1 (6 marks)

3.1 R14 500 is invested for 2.5 years at 9.6% per annum compounded quarterly. Calculate the final amount. [4]

3.2 An investment earns 11.2% per annum compounded half-yearly. How many compounding periods occur over 6.5 years? [2]

QUESTION 2 (4 marks)

4.1 Lebo wants R75 000 in 3 years. An account pays 8.4% per annum compounded monthly. Calculate the amount Lebo must invest today. [4]

QUESTION 5 (7 marks)

5.1 A laptop is purchased for R28 000 and depreciates by 15% per annum using the reducing balance method. Determine its value after 3 years. [4]

5.2 A piece of equipment is worth R96 000 after 4 years of straight-line depreciation at 10% per annum. Determine its original cost price. [3]

QUESTION 6 (4 marks)

6.1 Convert a nominal interest rate of 12.6% per annum compounded semi-annually to the equivalent annual effective interest rate. [2]

6.2 An account has an effective annual interest rate of 9.50% and compounds monthly. Determine the equivalent nominal annual interest rate. [2]

QUESTION 7 (4 marks)

7.1 Two banks offer the following options on a R20 000 investment for 2 years:
Bank A: 8.2% per annum compounded annually.
Bank B: 7.9% per annum compounded monthly.
Calculate the final amount from each bank. State the difference between the two final amounts. [4]